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RESEARCH ARTICLE

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Oncologists' preferences for first-line treatment of ALK-positive metastatic NSCLC in the United States: a discrete choice experiment

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ABSTRACT

Objective: Tradeoffs between efficacy, safety, and administration may influence decision-making for first-line (1L) treatments for ALK+ mNSCLC. This study explored heterogeneity in provider preferences.
Methods: A survey of US oncologists treating ≥1 ALK+ mNSCLC patient was conducted June–August 2024. Participants answered 12 questions comparing 1 L profiles characterized by eight treatment attributes for patients without brain metastases. Attributes included 3-year PFS, intracranial duration of response (iDOR), 16-month risk of edema, myalgia, or cognitive effects, grade ≥3 interstitial lung disease/pneumonitis, 2 L+ treatment options, and administration. Relative attribute importance (RAI) (0–100%) was calculated for aggregate efficacy and safety attributes. Latent class analysis (LCA) explored preference heterogeneity and differences in respondent characteristics.
Results: In total, 201 oncologists completed the discrete choice experiment. Oncologists emphasized the efficacy over safety (RAI = 47.0% vs 39.0%). LCA identified two classes, placing similar importance on efficacy (RAI = 44.5% vs 48.2%) and safety (RAI = 39% vs 41.2%), but preferring different levels in five attributes. Class 1 oncologists (61%) preferred 3-year PFS of 64% over 43–46%, while Class 2 oncologists (39%) did not differentiate. Conversely, Class 2 focused on iDOR.
Conclusion: We identified two oncologist groups with distinct preferences for treatment attributes. Given various 1 L ALK+ mNSCLC treatments, clinical decision-making balances multiple objectives.

PLAIN LANGUAGE SUMMARY

We surveyed 201 US-based oncologists to understand what characteristics of different treatment options for ALK+ metastatic non-small cell lung cancer impact clinical decisions. We asked about the role each treatment's efficacy, side effects, and administration play in treatment choices, using a Discrete Choice Experiment approach. We found that oncologists valued efficacy over a treatment's side effect profile and that there were two distinct groups of oncologists with respect to treatment preferences. While both groups placed similar value on efficacy outcomes, one group's clinical decisions were largely driven by a treatment's ability to slow disease progression, while the other group's clinical decision depended on a treatment's ability to protect the patient from developing brain metastases. This suggests that the treatment decisions for patients with ALK+ metastatic non-small cell lung cancer are complex and that physicians balance multiple-treatment objectives in their choice of therapy.

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KEYWORDS
ALK+ metastatic non-small-cell lung cancer; discrete choice experiment; provider preference; tyrosine kinase inhibitors; patient outcomes; decision making, treatment attributes

1. Introduction

Despite declining incidence rates, lung cancer remains the leading cause of cancer-related mortality among both men and women in the United States (US) [1,2]. Non-small-cell lung cancer (NSCLC) accounts for about 87% of all lung cancer diagnoses [3,4]. In the United States, around 2.6% of patients diagnosed with mNSCLC express the anaplastic lymphoma kinase (ALK+) genetic mutation [5]. Patients with ALK+ metastatic NSCLC tend to be younger at presentation and do not have a history of smoking compared to the overall lung cancer patient population [6]. Five-year median overall survival for patients with ALK+ mNSCLC is 60% [7]. ALK+ NSCLC is associated with significant economic burden, with an estimated direct medial cost of $22,000 per-patient per-month (PPPM; 2020 US Dollars) for patients on first-line (1 L) tyrosine kinase inhibitors (TKIs) [8].

In the US, the 1 L treatment landscape for ALK+ mNSCLC has transitioned from first-generation TKIs, mainly crizotinib, to the more efficacious second and third-generation TKIs: alectinib, brigatinib, and lorlatinib [9–11]. Next-generation TKIs demonstrate superior efficacy, reduced central nervous system (CNS) progression and improved overall survival (OS) compared to crizotinib [9–11]. However, these therapies offer differing trade-offs between efficacy endpoints, treatment-related adverse event (AE) profiles, and administration requirements. The National Comprehensive Cancer Network (NCCN) includes all three as preferred therapies in their recommendation for 1 L treatment [12,13]. However, the National Institute for Health and Care Excellence (NICE)

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Article highlights

- Survey of 201 US-based oncologists with experience treating patients with ALK+ mNSCLC, aiming to understand how differences in efficacy, safety, and administration influence treatment choice.
- Oncologists answered 12 questions comparing 1 L profiles characterized by eight treatment attributes for patients without brain metastases, including 3-year progression-free survival (PFS), intracranial duration of response (iDOR), 16-month risk of edema, myalgia or cognitive effects, grade ≥3 interstitial lung disease/pneumonitis, 2 L+ treatment options, and administration.
- In this discrete choice experiment survey, oncologists valued efficacy over safety when deciding between alternative treatment options.
- This study employed a latent class analysis to identify two distinct groups of oncologists, which place similar importance on efficacy and safety, but preferred different levels in five attributes. Specifically, oncologists in the first group (61%) preferred 3-year PFS of 64% over 43–46%, while oncologists in the second group (39%) did not differentiate between PFS-levels, but instead focused on iDOR.

recommends alectinib and brigatinib for 1 L treatment and specifies lorlatinib should only be used for patients previously treated with another second or third generations TKI [14–17]. The differing benefit-risk profiles of TKIs along with the shifting guideline landscape have created a need to understand provider preferences for the 1 L treatment of patients with ALK+ mNSCLC. This knowledge gap is critical to inform optimal therapeutic access and care provision, including guideline concordance and informed decision-making.

To date, no studies have evaluated provider preferences for the 1 L treatment of ALK+ mNSCLC [18–36]. The current discrete choice experiment (DCE) literature has primarily measured patient preferences for the treatment of mNSCLC with either chemotherapy or immunotherapy, or EGFR+ mNSCLC with TKIs [18,22,24,35]. Three studies have reported provider preferences for the treatment of mNSCLC with chemotherapy in the US and China, and for the maintenance therapy of patients with advanced non-squamous NSCLC [19,23,28]. None of these studies examined variations in provider preferences. Current treatment options enable patients with ALK+ mNSCLC to live longer than patients with non-driver mutation mNSCLC (median OS 48.5 vs 11.1 months) [37–39]. Therefore, patients and providers must manage toxicity profiles for longer duration, which may impact informed decision-making. In addition, the rarity of ALK+ mNSCLC suggests that some providers may have relatively little treatment experience with this population, which may influence their treatment preferences [40].

The objectives of the study were to : (1) quantify preferences for attributes of 1 L ALK+ mNSCLC treatment (e.g., progression-free survival, safety profile, mode of administration) among oncologists in the US, and (2) examine variations in oncologists' preferences for treatment attributes and the association of clinicians' demographic and clinical practice characteristics with different types of preferences.

2. Methods

2.1. Study design

A cross-sectional online discrete choice experiment (DCE) was administered between June and August 2024. DCEs are a commonly used method to assess preferences in healthcare [41]. Study participants completed a series of choice questions comparing two hypothetical treatments with varying levels of treatment attributes [40,42]. All study documents were reviewed and granted exemption from IRB oversight before recruitment by the Advarra Institutional Review Board (IRB) (protocol number 00078031).

2.2. Survey design

The DCE was designed following guidelines developed by the Professional Society for Health Economics and Outcomes Research (ISPOR) [43–45]. The DCE asked oncologists to consider *"A 52-year-old male patient, newly diagnosed with ALK+ metastatic NSCLC. This patient has no history of smoking and does not exhibit any brain metastases or mutations outside of ALK."* The DCE was calibrated to incorporate second and third-generation TKIs. The selection of the treatment attributes was informed by a targeted literature review of health technology assessments [14–16,46–52], DCEs in mNSCLC [18,19,22–26], clinical trial literature [9–11,53–56], prescribing information, and the expert opinion of two clinical advisors (author KM and Dr. Nathan Pennell of Cleveland Clinic). The final attributes included two efficacy attributes (probability of progression-free survival at 3-years, median intracranial duration of response among those with measurable brain metastases), four adverse event attributes (risk of any grade edema, risk of any grade myalgia, risk of severe (Grade ≥3) interstitial lung disease/pneumonitis, any cognitive effects), and two other attributes (mode and frequency of administration, subsequent treatment options) (Table 1). The median intracranial duration of response among those with measurable brain metastases was included because brain metastases are a common and critical site of disease progression in ALK+ mNSCLC [57]. This measure serves as an available proxy measure in assessing the impact of therapies on controlling intracranial disease. Mode and frequency of administration was included because it can increase pill burden and affect patient adherence. Subsequent treatment option allows providers to plan for subsequent treatment strategies in case of disease progression or tolerability issues with the 1 L therapy. The efficacy attribute levels were sourced within the ranges reported in the clinical trials of second- and third-generation TKIs [9,53,55,56,58–60], the adverse event attribute levels were sourced from their prescribing information, and scaled to 16-months, the mode of administration levels were also sourced from the prescribing information, and the subsequent treatment options allowed for potential treatment changes as recommended in the NCCN guidelines (Supplementary Exhibit 1) [11].

Figure 1 includes an example choice question. An opt-out or "neither" option was not included because foregoing 1 L treatment for ALK+ mNSCLC is not guideline recommended [54]. The attribute levels were varied according to a blocked fractional factorial experimental design constructed using AlgDesign package [10]. The D-efficiency criteria indicated that 36 choice questions were optimal [59]. To minimize survey burden, the experimental design was divided into three blocks of 12 choice questions and each respondent completed a randomly assigned block. The order the attributes were presented was randomized across respondents to minimize ordering biases [57,58].

Table 1. Attributes and levels.

	Attribute	Levels	Sources
Efficacy Attributes	Probability of progression-free survival at 3-years (95% confidence interval)	43% (CI = 34–51%) 46% (CI = 38–55%) 64% (CI = 55–71%)	[53–56]
	Median intracranial duration of response among those with measurable brain metastases (in months)	17 months 28 months > 36 months	[53–56]
Safety Attributes	Risk of any edema within the first 16 months of treatment*	12% 20% 54%	[57–59]
	Risk of any myalgias within the first 16 months of treatment*	14% 18% 21%	[57–59]
	Risk of severe (Grade ≥3) interstitial lung disease or pneumonitis within the first 16 months of treatment*	0.2% 0.6% 1.9%	[57–59]
	Risk of any cognitive effects within the first 16 months of treatment*	Less than 10% 20%	[57–59]
Other Attributes	Mode and frequency of administration	1 pill daily with or without food 4 pills twice a day with food	[53–59]
	Subsequent (2 L+) treatment option	Alternative 2nd/3rd generation TKI approved for treatment in 2 L Chemotherapy	Key opinion leader input (author KM and clinical advisor NP)

Note: *Median durations of treatment for relevant comparators differed. To control for the fact that clinical trials with longer follow-up time have more time at risk to see an adverse event occur, we scaled the available adverse event risks to 16 months. This scaling calculation assumes adverse events occur at a constant or linear rate during follow-up. 16 months was selected since it was the shortest duration and avoided inflating risk for any of the drugs. See Supplemental Exhibit #1 for an example of the calculation.

The survey instrument included questions related to respondents' demographics and practice characteristics. To assess the data quality, two additional non-experimental choice questions testing dominance and stability were included after each block (Supplementary Exhibit 2) [59]. The survey was pilot tested with five respondents, which led to further refinement of the survey instrument (Supplementary Exhibit 3). During the pilot interviews, participants completed the survey while sharing their screens via Zoom. The study team observed responses and recorded reactions and comments shared by the participants. Following survey completion, participants were asked about their perceptions of the survey, including length, comprehension, and adequacy of the experiment. Participants reported the survey to be straightforward and pointed out that the visual aids used to depict each treatment attribute were very helpful in interpreting the DCE choices.

2.3. Study population

A third-party online survey vendor (M3, US) with panels of clinician databases recruited the sample and administered the survey. The study included providers who self-reported a medical degree, physician assistant, or nurse practitioner; a primary oncology or hematology/oncology specialty; practiced at least some of their time in an outpatient setting in the US; had at least two-year post-training experience; spent at least 50% of their time providing direct patient care; treated at least one patient with ALK+ mNSCLC in the past year; and provided informed consent. The recruitment target of 200 oncology providers met accepted DCE sample size requirements [58]. Additional recruitment quotas required at least 50% of the sample to practice in community settings and no more than 35% from each of the four US census regions.

2.4. Statistical analysis

DCE data for the overall sample were analyzed using a random parameter logit model, which accounts for unobserved heterogeneity in preferences [12,57]. All attributes were treated as categorical variables, and effects coding was used to estimate a preference weight for each level [61,62]. A higher preference weight indicates that an attribute level is more preferred than a level of the same attribute with a lower preference weight. Statistical significance at the 0.05 level is present if the confidence intervals for any pair of levels within an attribute do not overlap [63]. Relative attribute importance (RAI) (range 0–100%) measured the contribution of each attribute to treatment choice relative to all the other attributes and conditional on the level ranges [64]. The higher the RAI, the more important an attribute. RAI aggregated at the efficacy, safety, and other attribute level was also reported. Confidence intervals for the RAIs were computed using a bootstrapping procedure with 10,000 replicates.

A secondary latent class logit model examined differences in preferences among subgroups of providers [65]. This analysis identified the number of classes (subgroups) with different preference patterns, the preference patterns in each class, and the provider characteristics of each identified class. The number of latent classes was selected based on model goodness-of-fit statistics and interpretability [65]. Preference weights and RAI were reported for each class. Summary statistics of the respondent demographics and practice characteristics in each class were reported. Unadjusted differences in the characteristics across the classes were examined using t-tests, chi-square tests, and Fisher exact tests. All analysis was conducted in RStudio Version 4.1.1.

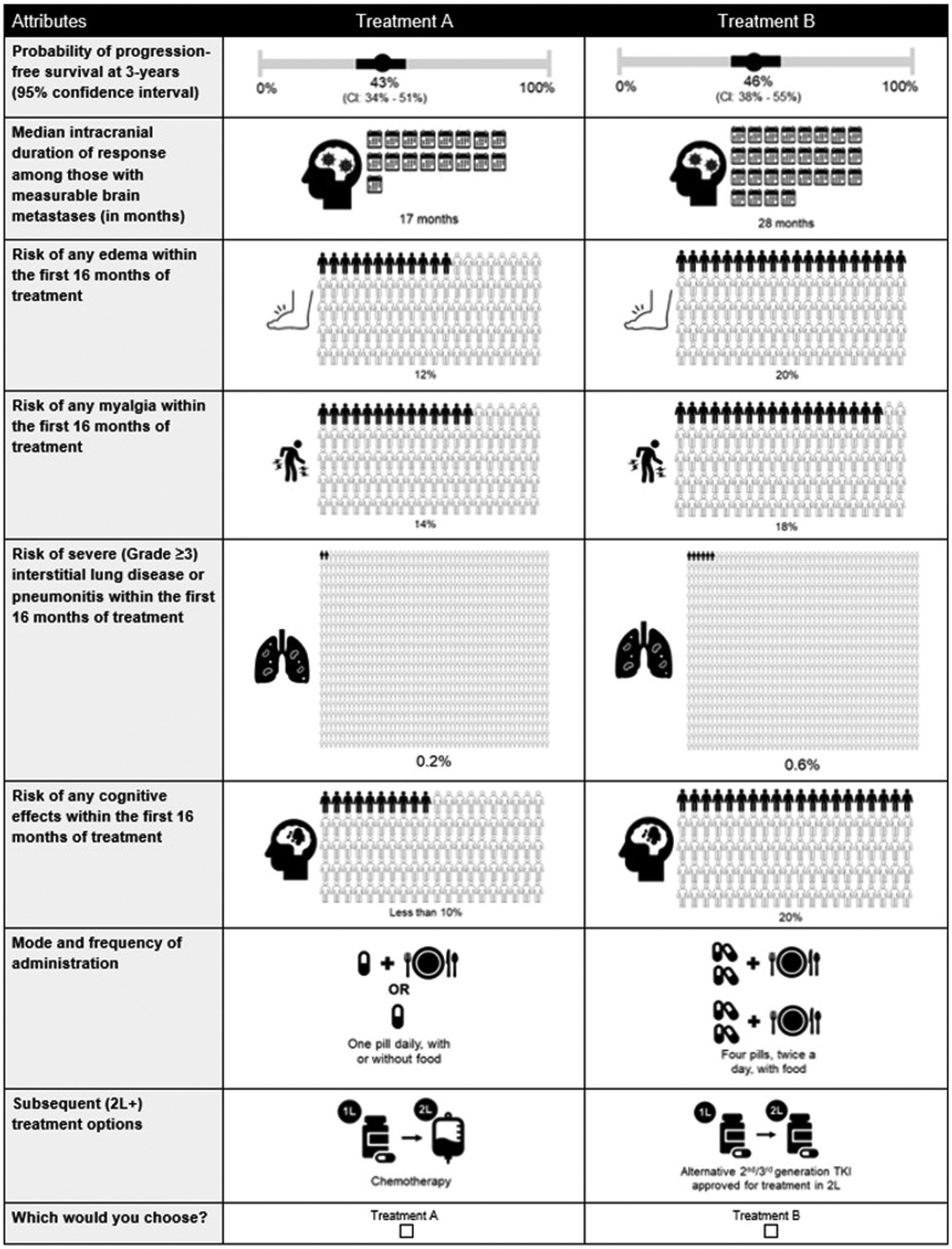

Figure 1. Example choice task.

3. Results

3.1. Respondent characteristics

In total, 319 participants responded to the study invitation. The screening questionnaire was started by 319 individuals of whom 118 were excluded because they did not complete it ($n = 25$), were not eligible ($n = 58$), or were flagged as a non-attentive respondent ($n = 35$). The final sample contained 201 oncology providers.

Survey respondents were predominantly male (68.2%), white or Caucasian (48.3%), not Hispanic, Latino or Spanish (74.6%), and specialized in hematology and oncology (51.2%). 54% of providers reported 15 years or fewer of practicing oncology post-residency and approximately two-thirds (64.7%) practiced in a community setting. Providers treated a median of 200 (IQR: 150,300) cancer patients per month and a median of 8 (IQR: 4, 20) ALK+ metastatic NSCLC patients in the past year (Table 2).

Table 2. Oncology provider characteristics.

	Full Sample (N = 201)	Class #1 (N = 121)	Class #2 (N = 79)	P-value Class #1 vs #2
Gender, n (%)				
Male	137 (68.2)	84 (68.9)	53 (67.1)	0.324
Female	48 (23.9)	31 (25.4)	17 (21.5)	
Prefer not to answer	16 (8.0)	7 (5.7)	9 (11.4)	
Race, n (%)				
American Indian or Alaska Native	0 (0.0)	0 (0.0)	0 (0.0)	0.414
Asian	48 (23.9)	30 (24.6)	18 (22.8)	
Black or African American	6 (3.0)	2 (1.6)	4 (5.1)	
Native Hawaiian or Other Pacific Islander	0 (0.0)	0 (0.0)	0 (0.0)	
White or Caucasian	97 (48.3)	61 (50.0)	36 (45.6)	
Two or more races	9 (4.5)	7 (5.7)	2 (2.5)	
Other	1 (0.5)	1 (0.8)	0 (0.0)	
Prefer not to answer	40 (19.9)	21 (17.2)	19 (24.1)	
Ethnicity, n (%)				
Hispanic, Latino, or Spanish origin	9 (4.5)	5 (4.1)	4 (5.1)	0.416
Not Hispanic, Latino or Spanish origin	150 (74.6)	95 (77.9)	55 (69.6)	
Prefer not to answer	42 (20.9)	22 (18.0)	20 (25.3)	
Primary medical specialty, n (%)				
Hematology and Oncology	103 (51.2)	51 (41.8)	52 (65.8)	0.001
Oncology	98 (48.8)	71 (58.2)	27 (34.2)	
Years practicing oncology post training, n (%)				
3–10 years	65 (32.3)	42 (34.4)	23 (29.1)	0.1
11–15 years	43 (21.4)	29 (23.8)	14 (17.7)	
16–20 years	41 (20.4)	18 (14.8)	23 (29.1)	
More than 20 years	52 (25.9)	33 (27.0)	19 (24.1)	
Time spent providing direct patient care, n (%)				
50–79%	10 (5.0)	6 (4.9)	4 (5.1)	1
80% or more	191 (95.0)	116 (95.1)	75 (94.9)	
Professional degree/license, n (%)				
Medical Degree (e.g., MD, MBBS, MBChB, Dr. Med, DO)	193 (96.0)	118 (96.7)	75 (94.9)	0.619
Physician Assistant (PA)	3 (1.5)	2 (1.6)	1 (1.3)	
Nurse Practitioner (NP)	5 (2.5)	2 (1.6)	3 (3.8)	
Practice type, n (%)				
Solo Practice	6 (3.0)	2 (1.6)	4 (5.1)	0.59
Partnership Practice	22 (10.9)	14 (11.5)	8 (10.1)	
Group Model Health Maintenance Organization (e.g., Kaiser Permanente)	4 (2.0)	3 (2.5)	1 (1.3)	
Government (e.g., VA, military, public health)	1 (0.5)	1 (0.8)	0 (0.0)	
University/Academic Medical Center	71 (35.3)	42 (34.4)	29 (36.7)	
Specialty Group Practice	35 (17.4)	18 (14.8)	17 (21.5)	
Multi-Specialty Group Practice	28 (13.9)	19 (15.6)	9 (11.4)	
Hospital-Based Practice	34 (16.9)	23 (18.9)	11 (13.9)	
Practice setting, n (%)				
Academic	71 (35.3)	42 (34.4)	29 (36.7)	0.857
Community	130 (64.7)	80 (65.6)	50 (63.3)	
Practice location, n (%)				
Major metropolitan area (population > 500,000)	70 (34.8)	45 (36.9)	25 (31.6)	0.821
Urban area (population 100,000 – 500,000)	52 (25.9)	30 (24.6)	22 (27.8)	
Suburb of a large city (population > 100,000)	56 (27.9)	34 (27.9)	22 (27.8)	
Small city (population 30,000 – 100,000)	16 (8.0)	10 (8.2)	6 (7.6)	
Rural or small town (population < 30,000)	7 (3.5)	3 (2.5)	4 (5.1)	
Cancer patient volume per month				
Mean (SD)	241.29 (143.61)	214.00 (124.02)	283.43 (161.42)	0.001
Median [IQR]	200 [150, 300]	200 [129.25, 278.75]	300 [155, 375]	0.001
NSCLC patient volume per month				
Mean (SD)	62.87 (60.09)	53.57 (35.69)	77.23 (83.29)	0.006
Median [IQR]	50 [30, 75]	45 [30, 70]	50 [34.5, 80]	0.031
Metastatic NSCLC patient volume per month				
Mean (SD)	45.27 (46.81)	37.89 (25.59)	56.67 (66.22)	0.005
Median [IQR]	33 [20, 50]	30 [20, 50]	40 [20, 60]	0.089
ALK+ metastatic NSCLC patient volume in the past year				
Mean (SD)	31.77 (89.60)	25.22 (76.03)	41.87 (107.00)	0.199
Median [IQR]	8 [4, 20]	7 [4, 15]	10 [5, 30]	0.016

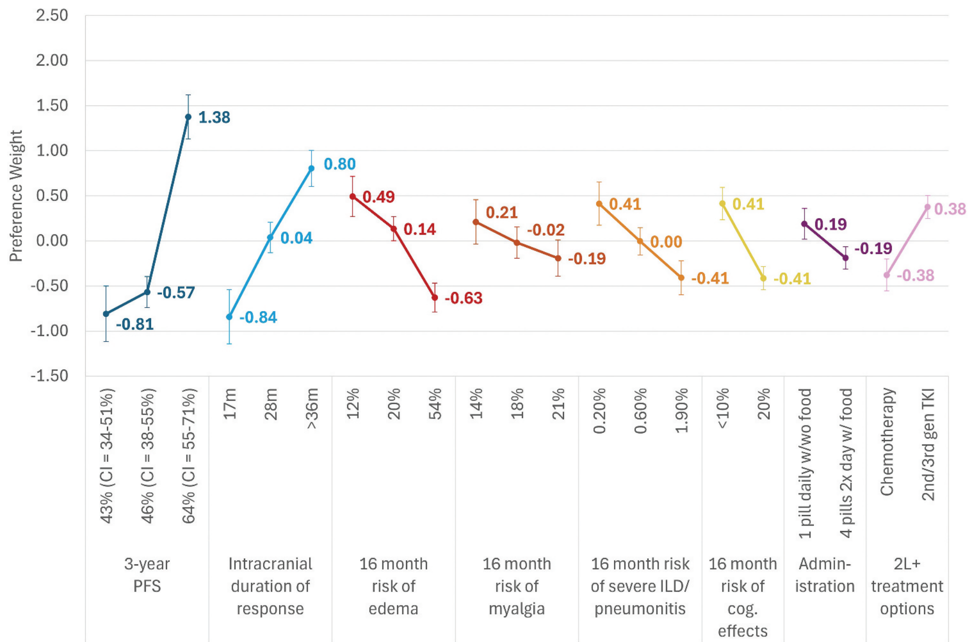

Figure 2. Preference weights for ALK+ mNSCLC 1 L treatment attributes (N = 201).

Data quality was good with most participants passing the data quality checks (dominance test = 92%; stability test = 83%).

3.2. Preference weights and relative attribute importance

There were statistically significant differences in the preference weights for many of the attribute levels (Figure 2(a)). Oncologists preferred a 3-year PFS of 64% over 46% or 43%, and a median intracranial duration of response of >36 months over 28 months, which was preferred over 17 months. Among the safety attributes, oncology providers preferred a 12% or 20% 16-month risk of edema over 54%, a 0.2% or 0.6% 16-month risk of severe ILD/pneumonitis over 1.9%, and <10% 16-month risk of cognitive effects over 20%. There were no statistically significant differences between any of the levels of 16-month risk of myalgia (14%, 18%, 21%). Finally, the respondents preferred treatments given via one pill daily with or without food vs the four pill two times daily with food alternative and those followed by a 2nd/3rd generation TKI in 2 L+ compared to chemotherapy (Supplementary Exhibit 4).

The RAIs indicated that all attributes included in the DCE factored into treatment decision-making (Supplementary Exhibit 5). Overall, oncology providers placed greater importance on the efficacy attributes (Aggregated RAI = 47.0%) than on the four safety attributes (Aggregated RAI = 39.0%) or the other attributes, administration and 2 L+ treatment options (Aggregated RAI = 13.9%). The 3-year PFS had the strongest influence on treatment choice (RAI = 26.8%) followed by intracranial duration of response (RAI = 20.2%). A 16-month risk of any edema was the third most important attribute and top safety attribute (RAI = 13.8%) followed by 16-month risk of any cognitive effects (RAI = 10.2%) and 16-month risk of severe ILD/pneumonitis (RAI = 10.1%). 2 L+ treatment options (RAI = 9.3%) was the sixth most important attribute. Finally, 16-month risk of any myalgia (RAI = 4.9%) and mode/frequency of administration (RAI = 4.6%) were the least important.

3.3. Preference heterogeneity

Two classes of preferences were identified in the sample: Class 1 included N = 122 (61%) of the providers and Class 2 had N = 79 (39%) (Supplementary Exhibit 6). Overall, providers in Class 1 were more likely to see lower volumes of cancer patients, NSCLC patients, mNSCLC patients, and ALK+ mNSCLC patients than providers in Class 2. In particular, Class 1 (61%) treated fewer total cancer patients per month than Class 2 (39%) [median = 200 vs 300; $p < 0.001$] and fewer ALK+ mNSCLC patients per year [median = 7 vs 10; $p = 0.016$]. In addition, fewer providers in Class 1 vs Class 2 reported hematology and oncology as their primary specialty (41.8% vs 65.8%; $p = 0.001$). There were no statistically significant differences in providers' age, gender, ethnicity, years practicing oncology, time spent providing direct patient care, and the practice type, urbanicity, and academic vs community setting between the classes (Supplementary Exhibit 7).

The two classes exhibited different patterns of statistically preferred levels in five of the eight attributes (Figure 3(a,b)). For 3-year PFS, Class 1 preferred 64% over 46% or 43%, while there were no statistically significant differences between any of the levels for Class 2. For intracranial duration of response, Class 1 preferred >36 or 28 months over 17 months, while Class 2 preferred >36 months over 28 months, which was preferred over 17 months. For 16-month risk of any edema, Class 1 preferred 12% over 20%, which was preferred over 54%, while Class 2 preferred 12% or 20% over 54%. In terms of 16-month risk of severe ILD/ pneumonitis, Class 1 preferred 0.2% or 0.6% over 1.9% while Class 2 preferred 0.2% over 0.6% or 1.9%. Finally, Class 1

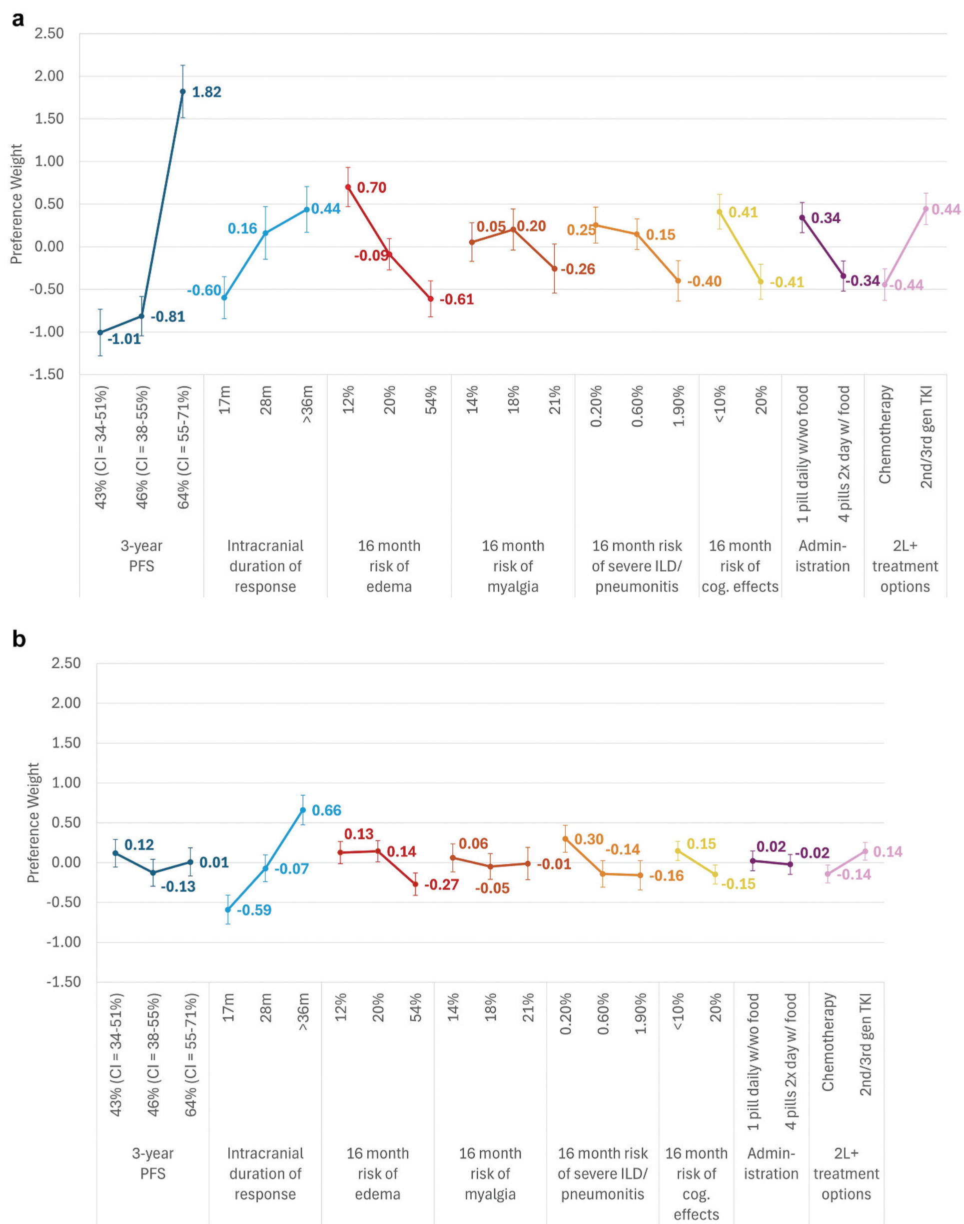

Figure 3. Preference weights for ALK+ mNSCLC 1 L treatment attributes, by latent class: a) preference weights for class 1 (N = 122), b) preference weights for class 2 (N = 79).

preferred one pill daily (with or without food) vs the four pills two times a day (with food) alternative, while there were no statistically significant differences between any of the levels for Class 2. For the remaining three attributes, both classes preferred <10% 16-month risk of cognitive effects and the option for a 2nd/3rd generation TKI in 2 L+, and neither class exhibited statistically significant differences between any of the levels of 16-month risk of myalgia (Supplementary Exhibit 8).

Class 1 placed the highest importance on the efficacy attributes (Aggregated RAI = 44.5%), followed by the safety attributes (Aggregated RAI = 37.3%) and the other attributes (Aggregated RAI = 18.1%). Class 2 had the same ordering but with slightly different values (efficacy RAI = 48.2%, safety RAI = 41.2%, and other RAI = 10.6%) (Supplementary Exhibit 9). All attributes included in the DCE factored into treatment decision-making in both classes. However, Class 1 placed more importance on 3-year PFS than Class 2 (RAI = 32.6%, CI=[27.6–37.8%] vs 7.9%, CI=[2.0–15.1%]), while Class 2 placed more importance on intracranial duration of response than Class 1 (RAI = 40.3%, CI=[28.6–45.0%] vs 11.9%, CI=[8.1–16.0%]). There were no statistically significant differences in the RAI of the remaining attributes between the classes. However, the second and third most important attributes were 16-month risk of edema (RAI = 15.1%) and 2 L+ treatment options (RAI = 10.2%) for Class 1, and 16-month risk of severe ILD/ pneumonitis (14.7%) and 16-month risk of edema (13.4%) for Class 2.

4. Discussion

This study conducted in the US found that oncologists placed slightly more importance on the efficacy than the safety attributes (RAI = 47.0% vs 39.0%) for the first-line treatment of ALK+ NSCLC, and relatively less importance on the mode of administration and 2 L+ treatment options (RAI = 13.9%). Oncologists preferred the highest levels of both efficacy attributes (e.g., 3-year PFS and median intracranial duration of response) and the lowest levels for three of the four safety attributes (16-month risk of edema, 16-month risk of severe ILD/pneumonitis, and 16-month risk of cognitive effects). There were no statistically significant differences in their preferences between any of the levels of 16-month risk of myalgia. These results suggest that oncologists had preferences that distinguished between many of the levels in the treatment attributes except for 16-month risk of myalgia. To our knowledge, this is the first study examining preferences for attributes of 1 L ALK+ mNSCLC treatment. As such, it included different attributes, specifically intracranial duration of response and the four safety attributes, than found in the prior preference studies of the treatment of mNSCLC [18–27,44,66–72]. However, consistent with the prior preference literature, our study found that oncologists placed more importance on the efficacy than on the safety attributes, although the specific attributes differed [28–30].

This study further identified two groups (classes) of oncologists that placed similar importance on the efficacy (RAI = 44.5% vs 48.2%) and the safety attributes (RAI = 39% vs 41.2%), but preferred different levels of the efficacy attributes (3-year PFS, median intracranial duration of response), of two of the safety attributes (16-month risk of edema, 16-month risk of severe ILD/pneumonitis), and of mode of administration. For 3-year PFS, Class 1 preferred 64% over 46% or 43%, while Class 2 did not distinguish between different levels. In contrast, Class 2 focused on intracranial duration of response as their main efficacy outcome. These results indicate the two classes of oncologists preferred different approaches to balancing the therapeutic benefit-risk complexities. In addition, Class 1 had lower volumes of cancer patients, NSCLC patients, mNSCLC patients, and ALK+ mNSCLC patients than Class 2 suggesting an association between patient volume and oncologists' preferences for treatment of ALK+ mNSCLC.

The results from this study have numerous implications for general practice. Our finding that intracranial duration of response was the second most important attribute overall, even for the treatment of an ALK+ mNSCLC patient without any brain metastases, highlights the value oncologists placed on mitigating the risk of progression to the brain, particularly since up to 75% of ALK+ mNSCLC patients will develop brain metastases during the course of their disease [31–33]. The pilot tests supported the perception of this attribute as a proxy for preventing cranial involvement given the lack of a comparable clinical trial endpoint measuring the time to progression in the brain and reiterated the value placed on a potential "ability to prevent new brain metastasis from developing." Our finding that 16-month risk of any edema was the most important safety attribute to oncologists may reflect the belief that this is a problematic adverse event for patients with little therapeutic recourse for oncologists besides discontinuing the TKI. In contrast, cognitive effects (RAI = 10.2%) and ILD/pneumonitis (RAI = 10.2%) had lower relative attribute importance in spite of the potential for severe or refractory effects warranting interruption or reduction as well as escalation in care with additional medications and specialty referrals [34,35]. Oncologists also value 2 L+ treatment options (RAI = 9.3%), with a preference for a 2nd/3rd generation TKI over chemotherapy, suggesting treatment sequencing is also a relevant factor when considering 1 L options. Finally, although oncologists placed the least importance on the administration, they preferred one pill daily without food over four pills two times daily with food potentially due to concerns about pill burden and long-term adherence.

In addition, the current study contributes to the understanding of heterogeneity in oncologists' treatment preferences for the same standard ALK+ mNSCLC patient by identifying two groups (classes) of oncologists with distinct preferences. We found that although both classes balanced efficacy and safety, Class 1 placed more importance on 3-year PFS and Class 2 on intracranial duration of response. Class 1 preferred a 3-year PFS of 64% over 46% or 43%, suggestive of preferences similar to the findings of those for the treatment of mNSCLC [19,28,36], while Class 2 did not distinguish between the different levels of PFS. In contrast, Class 2 preferred the highest levels of attributes that avoided some of the worst non-survival outcomes associated the disease (specifically intracranial duration of response) or treatment (risk of severe ILD/pneumonitis). Compared to Class 2, Class 1 exhibited larger differences in preferences between all the attribute levels with a relative focus on 3-year PFS, while Class 2 exhibited fewer differences between attribute levels with a relative focus on intracranial duration of response. Given that Class 2 oncologists may represent more specialized experts, there is an opportunity to highlight their highest attributes, specifically brain protection as well as ILD mitigation, and create decision-support tools or educational material to ensure all oncologists are educated in these high-impact, long-term disease risks, particularly when survival is improved among all agents.

Our finding that Class 1 treated fewer total cancer patients and ALK+ mNSCLC may have some potential explanations. Oncologists with higher levels of direct experience treating ALK+ patients and using the various drugs may have a more nuanced/balanced risk-benefit approach to drug selection, while those with less experience are more influenced by published efficacy results. Alternatively, oncologists who treat fewer total cancer patients may be more likely to practice in smaller community practices or have other practice characteristics (beside academic vs community) that were not measured in this survey. We note that not all low volume providers are in Class 1 and conversely not all high volume providers are in Class 2, suggesting that there are other factors driving class affiliation that are not included in this study. Future studies should explore how oncologist's experience managing brain metastases may influence preferences. These results highlight how general differences in oncologist's preferences for treating ALK+ mNSCLC patients may be driven by patient volume, highlighting the importance of communicating to patients the multiple aspects of achieving the best possible tumor response and managing toxicities.

None of the oncologist DCEs in mNSCLC [23,73,74] and few DCE studies in oncology have examined differences in preferences among subgroups of oncologists using a latent class analysis [75]. However, prior DCE studies of oncologists who treat metastatic urothelial carcinoma in Europe, and oncologists who treat metastatic colorectal cancer in the US have found that different classes prefer different trade-offs between the efficacy and safety attributes, with approximately 30% of oncologists placing the greatest importance on safety [76]. Further, the study in Europe found that oncologists who treated lower volumes of patients were more likely to place the highest importance on the safety vs the efficacy attributes [77]. In contrast, this study found two classes of oncologists that preferred different treatment approaches to achieve a similar trade-off between efficacy and safety. Further, oncologists treating higher volumes of patients placed greater importance intracranial duration of response compared to 3-year PFS. These findings add to the growing literature suggesting that oncologists' practice characteristics may influence their treatment preferences.

4.1. Limitations

The results of this study may be subject to information bias and selection bias resulting from the convenience, panel-based recruitment methods and study design. As with all survey-based research, respondents volunteer to be part of this study, and thus self-selection bias may be present. The web-based nature of the survey is also likely to result in a respondent sample with adequate access to and ability to use technology. This means our approach possibly under-represents oncologists practicing in rural or underserved areas, although we tried to counteract this risk by over-sampling respondents from community-based practice settings. Survey recruitment based on a convenience sample limits the opportunity to calculate nonresponse bias. All respondent characteristics were self-reported. Further, the DCE questions require respondents to make hypothetical choices by evaluating the trade-offs between attributes. These hypothetical scenarios do not fully reflect the complex nature of ALK+ mNSCLC, its treatment, and characteristics of individual patients; thus, results may not predict actual decisions made in clinical settings. In particular, choices are presented for a patient without brain metastases, and clinical decision-making might differ for patients with brain metastases or other clinical characteristics not included in the experiment. Furthermore, although use of iDOR as a proxy for CNS protection was justified in this study given the lack of a consistent measure across clinical trials, it remains unclear how preferences may change with the inclusion of a specific CNS protection attribute. Similarly, the AEs considered in this DCE are key differentiators between commonly used 1 L treatments in ALK+ mNSCLC, but the DCE does not consider other important AEs that might impact decision-making, such as chronic side effects. While the results of this DCE reflect the views of oncologists, the voice of the patient is not captured and may differ, emphasizing the ongoing need for informed decision-making encompassing outcome and toxicity data. Although NICE, NCCN, and ESMO are currently aligned in not preferring one 1 L treatment over a collection of options, this provides an opportunity to harmonize guideline language to base treatment choice on both informed and shared decision-making, similar to that noted in The National Institute for Health and Care Excellence (NICE) guidelines.

5. Conclusions

We found two groups of oncologists, which varied by patient volume, with different preferences for efficacy (3-year PFS, in particular), safety, and other attribute levels. With various 1 L ALK+ mNSCLC treatments available, oncologists may balance multiple-treatment objectives when making informed treatment decisions with patients.

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Author contributions

Kathryn F. Mileham: Methodology, Validation, Writing – Original Draft, Writing – Review & Editing; *Laura Panattoni:* Conceptualization, Methodology, Validation, Formal Analysis, Investigation, Supervision, Project administration, Writing – Original Draft, Writing – Review & Editing, Funding acquisition; *Grace Gahlon:* Methodology, Validation, Formal Analysis, Data Curation, Visualization, Investigation, Writing – Original Draft, Writing – Review & Editing; *Pedro Labisa:* Conceptualization, Methodology, Validation, Writing – Original Draft, Writing – Review & Editing; *Nicole Bariahtaris:* Methodology, Validation, Formal Analysis, Data Curation, Visualization, Investigation, Writing – Original Draft, Writing – Review & Editing, Funding acquisition; *Marlon Graf:* Conceptualization, Methodology, Validation, Formal Analysis, Investigation, Supervision, Project administration, Writing – Original Draft, Writing – Review & Editing, Funding acquisition; *Christopher Danes:* Methodology, Validation, Writing – Original Draft, Writing – Review & Editing; *Luis Hernandez:* Conceptualization, Methodology, Validation, Supervision, Project administration, Writing – Original Draft, Writing – Review & Editing, Funding acquisition.

Disclosure statement

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Ethical declaration

All study documents were reviewed and granted exemption from IRB oversight before recruitment by the Advarra Institutional Review Board (IRB) (protocol number 00078031). Prior to beginning the survey, participants were presented with an information statement detailing study purpose as well as potential risks and benefits. In-lieu of a signature, participants indicated their consent using a radio button (Options “Yes” and “No”) and were then able to proceed to the survey if they answered “Yes” and omitted if they answered “No.”

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Data sharing statement

The data that support the findings of this study are available from the corresponding author, LH, upon reasonable request.

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